

Quick Start Guide - Hospital Appointment Management System



Running the Application

Prerequisites

- Java 17 or higher
- Maven 3.9+
- MySQL 8.0+

Database Setup

```
# 1. Login to MySQL
```

```
mysql -u root -p
```

```
# 2. Run the database creation script
```

```
source create_database.sql
```

```
# OR manually create:
```

```
CREATE DATABASE hospital_mvp_new;
```

```
CREATE USER 'hospital_user'@'localhost' IDENTIFIED BY 'password123';
```

```
GRANT ALL PRIVILEGES ON hospital_mvp_new.* TO 'hospital_user'@'localhost'
```

```
FLUSH PRIVILEGES;
```

Start Application

```
# Method 1: Using Maven
```

```
mvn spring-boot:run
```

```
# Method 2: Build and run JAR
```

```
mvn clean package -DskipTests
```

```
java -jar target/hospital-mvp-0.0.1-SNAPSHOT.jar
```

Visit: <http://localhost:8080>

Test Credentials

Patients

- **Email:** `patient@example.com`
- **Password:** `password123`

Doctors (10 accounts)

Email	Specialty	Password
doctor1@example.com	General Medicine	password123
doctor2@example.com	Cardiology	password123
doctor3@example.com	Dermatology	password123
doctor4@example.com	Pediatrics	password123
doctor5@example.com	Orthopedics	password123
doctor6@example.com	Neurology	password123
doctor7@example.com	ENT	password123
doctor8@example.com	Gynecology	password123
doctor9@example.com	Psychiatry	password123
doctor10@example.com	Oncology	password123

Admin

- **Email:** `admin@example.com`
- **Password:** `password123`



User Workflows

Patient Workflow

1. Sign Up

- Click "Login / Sign Up"

- Switch to "Sign Up" tab
- Enter: Name, Email, Phone, Password
- Click "Create Account"

2. Login

- Email: Your registered email
- Password: Your password
- Role: Patient
- Click "Login"

3. Book Appointment

- Search doctors by name or specialty
- Filter by specialty dropdown
- Click on a doctor card
- Select a date
- Click "Load Slots"
- Click on a **green** slot (available)
- Confirm booking

4. View Profile

- Click "My Profile" in navbar
- See upcoming appointments
- See past appointments
- View medical reports

Doctor Workflow

1. Login

- Email: `doctor1@example.com`
- Password: `password123`
- Role: Doctor
- Click "Login"

2. View Dashboard

- See statistics (total, today, upcoming, completed)
- View today's appointments
- View all appointments

3. Manage Slots

- Select a date
- Click "Load Slots"
- View your schedule (green = available, red = booked)

4. Add Medical Notes

- Find an appointment
- Click to add remarks/prescription
- Mark as complete

Admin Workflow

1. Login

- Email: `admin@example.com`
- Password: `password123`
- Role: Admin
- Click "Login"

2. View Analytics

- Total patients
 - Total doctors
 - Total appointments
 - Today's appointments
-



UI Features

Color Coding

-  **Green Slots:** Available (you can book)
-  **Red Slots:** Booked (disabled)
- **Blue Buttons:** Primary actions
- **Gray Buttons:** Secondary actions

Navigation

- **Home:** Return to landing page (stays on home if clicked from dashboard)
- **My Profile:** View your profile and appointments
- **Logout:** Clear session and logout

Session Management

- Login persists for **30 days**
 - Works across browser tabs
 - Stored in both localStorage and cookies
 - Auto-redirects to dashboard if already logged in
-



Troubleshooting

Issue: "Access Denied" on /profile

Solution: Build succeeded - restart the application

```
mvn clean spring-boot:run
```

Issue: Slots not showing colors

Solution: Hard refresh the browser

- Chrome/Firefox: `Ctrl + Shift + R` (Windows/Linux)
- Mac: `Cmd + Shift + R`

Issue: Doctor appointments not loading

Solution:

1. Make sure you're logged in as a doctor
2. Check browser console for errors (F12)
3. Verify doctor ID is being extracted from token

Issue: "Connection refused" to MySQL

Solution:

1. Make sure MySQL is running
2. Check credentials in `application.properties`
3. Verify database `hospital_mvp_new` exists

Issue: Compilation errors

Solution:

```
mvn clean compile
# If errors persist, check Java version
java -version # Should be 17+
```

Project Structure

```
java-project/
├─ src/
│   ├─ main/
│   │   └─ java/com/example/hospitalmvp/
│   │       ├─ controller/      # REST API endpoints
│   │       ├─ service/        # Business logic
│   │       ├─ repository/     # Database access
│   │       ├─ entity/         # JPA entities
│   │       ├─ security/       # JWT & auth
│   │       └─ dto/            # Data transfer objects
│   │   └─ resources/
│   │       ├─ static/         # CSS, JS
│   │       ├─ templates/     # HTML pages
│   │       └─ application.properties
│   └─ test/                  # Test files
├─ pom.xml                   # Maven dependencies
├─ REPORT.md                 # Detailed project report
├─ FIXES_SUMMARY.md          # All fixes documented
├─ DOCTOR_CREDENTIALS.md     # Login credentials
└─ QUICK_START.md            # This file
```

Configuration

Database Configuration

File: `src/main/resources/application.properties`

```
spring.datasource.url=jdbc:mysql://localhost:3306/hospital_mvp_new
spring.datasource.username=hospital_user
spring.datasource.password=password123
```

JWT Configuration

- **Token Expiry:** 7 days (default) or 30 days (remember me)
- **Secret Key:** Generated automatically
- **Algorithm:** HS256

Server Configuration

- **Port:** 8080
 - **Context Path:** /
 - **Session:** Stateless (JWT-based)
-



API Endpoints

Public Endpoints (No Auth Required)

GET /api/public/doctors - List all doctors
GET /api/public/doctors/{id}/slots?date=YYYY-MM-DD - Get doctor slots
POST /api/auth/login - Login
POST /api/auth/register/patient - Register patient

Patient Endpoints (Requires PATIENT role)

GET /api/patient/appointments/{patientId} - My appointments
POST /api/patient/appointments - Book appointment
DELETE /api/patient/appointments/{id} - Cancel appointment

Doctor Endpoints (Requires DOCTOR role)

GET /api/doctor/appointments/{doctorId} - My appointments
GET /api/doctor/slots/{doctorId} - My slots
PATCH /api/doctor/slots/{id}/availability - Toggle slot
PATCH /api/doctor/appointments/{id}/remarks - Add notes/prescription

Admin Endpoints (Requires ADMIN role)

GET /api/admin/patients - All patients
GET /api/admin/doctors - All doctors

Testing Checklist

- ☐ Patient can register
 - ☐ Patient can login
 - ☐ Patient can search doctors
 - ☐ Patient can see green (available) and red (booked) slots
 - ☐ Patient can book an appointment
 - ☐ Patient can view profile
 - ☐ Patient can cancel appointment
 - ☐ Doctor can login
 - ☐ Doctor can see appointments
 - ☐ Doctor can view slots
 - ☐ Doctor can add prescription
 - ☐ Admin can login
 - ☐ Admin can see analytics
 - ☐ Session persists across navigation
 - ☐ Logout clears session
-

Getting Help

Error Messages

All API errors return JSON:

```
{
  "message": "Error description here"
}
```

Browser Console

Press **F12** to open developer tools and check:

- **Console tab:** JavaScript errors
- **Network tab:** API request/response
- **Application tab:** LocalStorage and cookies

Logs

Check application logs for backend errors:

```
# If running with Maven
# Logs appear in terminal

# If running as JAR
java -jar target/hospital-mvp-0.0.1-SNAPSHOT.jar > app.log 2>&1
```



Documentation

- **REPORT.md**: Complete technical documentation (75+ pages)
 - **FIXES_SUMMARY.md**: All bugs fixed and how
 - **DOCTOR_CREDENTIALS.md**: All login credentials + hero image guide
 - **WORKFLOW_IMPROVEMENTS.md**: Feature improvements made
-



Build Status

```
[INFO] BUILD SUCCESS
[INFO] Total time: 11.128 s
```

All systems ready! 🎉

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